

Spinal Reformation in Scoliosis CT: Unlocking Reliable Morphometric Assessment for Vertebral Disease Detection

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1. Challenges in AI-based Spine Disease Detection for Scoliosis Patients

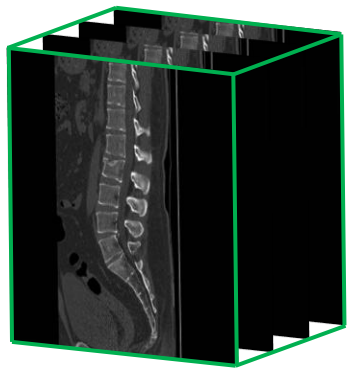
- Most AI models are trained on normal or mildly deformed spines, resulting in poor performance in severe scoliosis cases.
- Complex spinal curvature and patient-specific variability make automatic vertebrae localization and segmentation in CT highly challenging.
- In sagittal CT images of scoliosis patients, AI-based VCF (Vertebral Compression Fracture) detection is limited due to incomplete spine visibility within a single image slice.

[Ref]: Saeed, M. U., Dikaio, N., Dastgir, A., Ali, G., Hamid, M., & Hajje, F. (2023). An automated deep learning approach for spine segmentation and vertebrae recognition using computed tomography images. *Diagnostics*, 13(16), 2658. <https://doi.org/10.3390/diagnostics13162658>

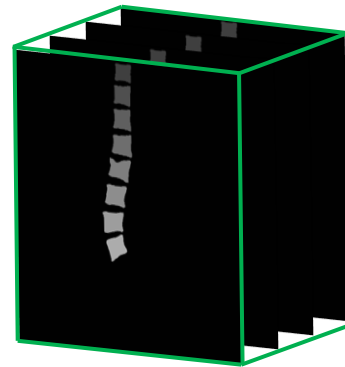
- To generate reformatted and anatomically corrected images using 3D coordinate-based processing, enabling clearer visualization and facilitating more accurate downstream analysis in scoliosis patients.

Overview of Spinal Reformation in CT

Using DL-based Spine Segmentation Model

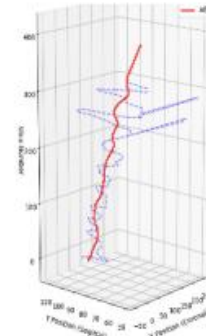


Input
CT Volume Data

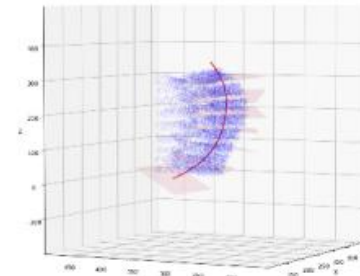


Output
Mask Volume Data

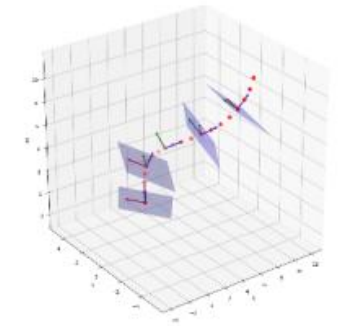
(a) 3D Visualization of Spine Centerline



(b) Spine Visualization with Real Data



(c) Spine Reformation with Coordinate Systems



Find Spine Centerline and
Reformation



(a) Before Scoliosis Reformat

(b) After Scoliosis Reformat



Final Output

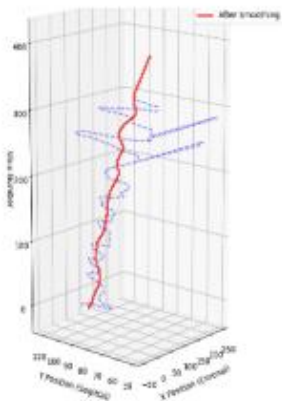
Dataset Description and Evaluation Methods

- A total of 34 cases of scoliosis patients were included.
- The measurements were performed by two experts in spine imaging engineering.
- The test dataset consisted of 24 cases with mild scoliosis (Cobb angle $\leq 25^\circ$) and 10 cases with moderate to severe scoliosis (Cobb angle $> 25^\circ$).
- **Qualitative Evaluation:** Reformatted axial and sagittal images were visually assessed to determine anatomical plausibility and image clarity improvements.
- **Quantitative Evaluation:** Cobb angles were measured before and after reformation. A successful scoliosis correction is indicated by a noticeable reduction in Cobb angle, reflecting improved spinal alignment.

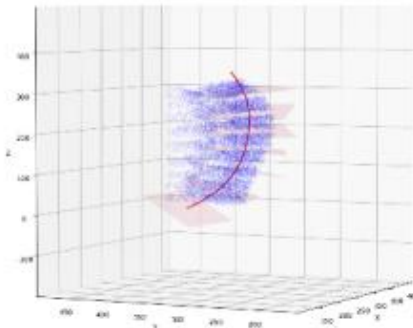
Spine Centerline-Based Reformation Process

- Extract the 3D spine centerline utilizing spine mask volume data.
- Perform smoothing operations on the extracted centerline to remove outliers and irregularities.
- Conduct reformation using the smoothed centerline.
- The reformation process follows equations (1) and (2).

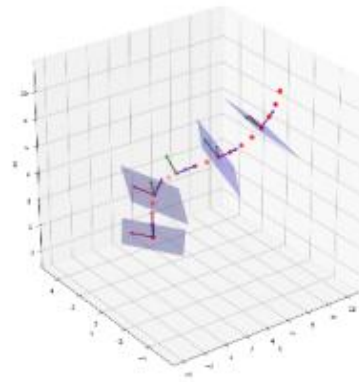
(a) 3D Visualization of Spine Centerline



(b) Spine Visualization with Real Data



(c) Spine Reformation with Coordinate Systems



$$x_k = x_1 + \frac{j}{10}(x_2 - x_1), \quad z_k = z_1 + \frac{j}{10}(z_2 - z_1), \quad w = j * 0.1 \quad (1)$$

$$I_k(y, z) = (1 - w) * I(x_1, y, z) + w * I(x_2, y, z), \quad z \in [z_{k-1}, z_k] \quad (2)$$

Qualitative Evaluation

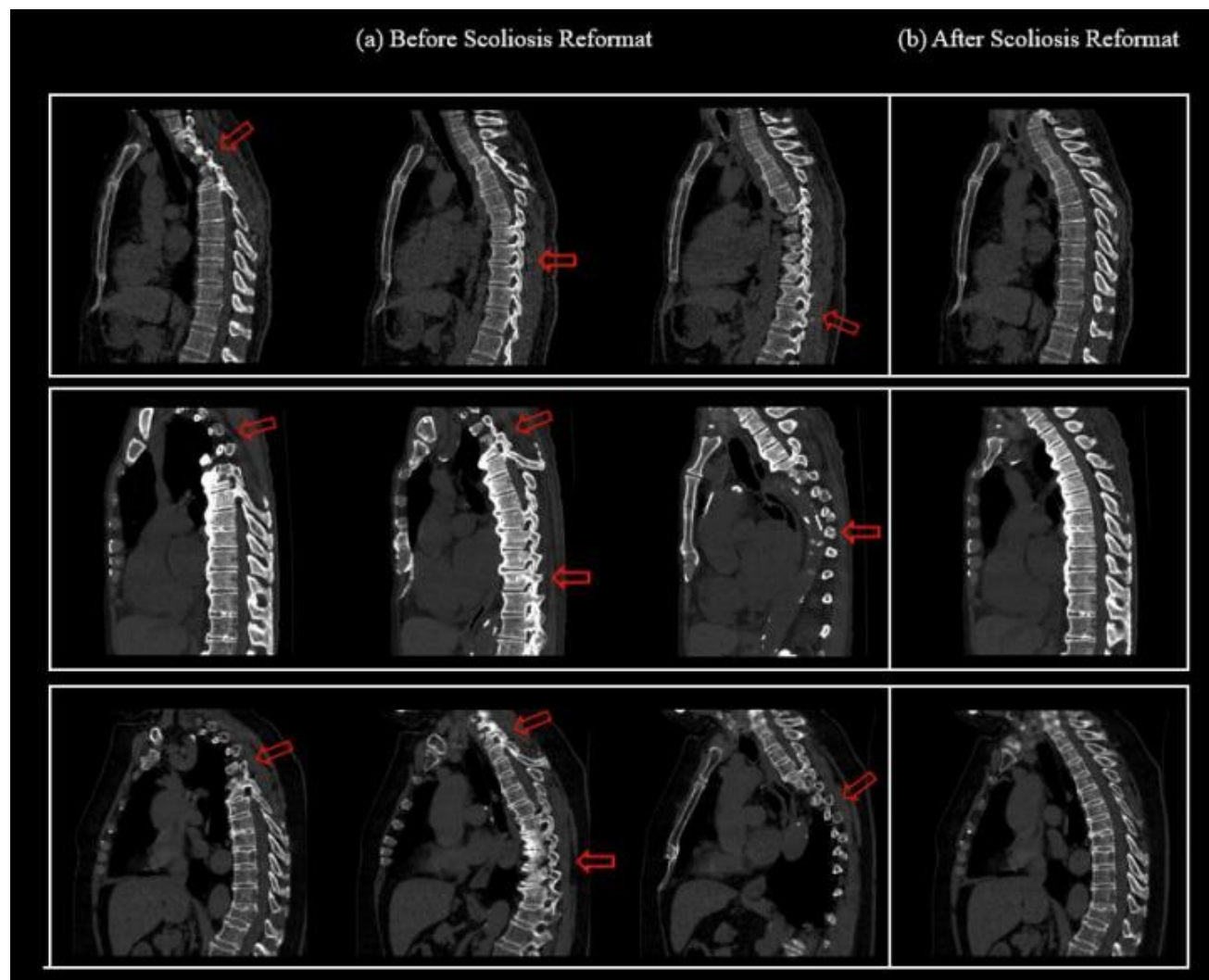


Fig. 1. Visualization of the scoliosis reformation outcomes. (a) shows the original sagittal CT image before reformation, while (b) presents the image after applying the proposed reformation method. Red arrows highlight areas where the spine is obscured due to scoliosis.

Qualitative Evaluation

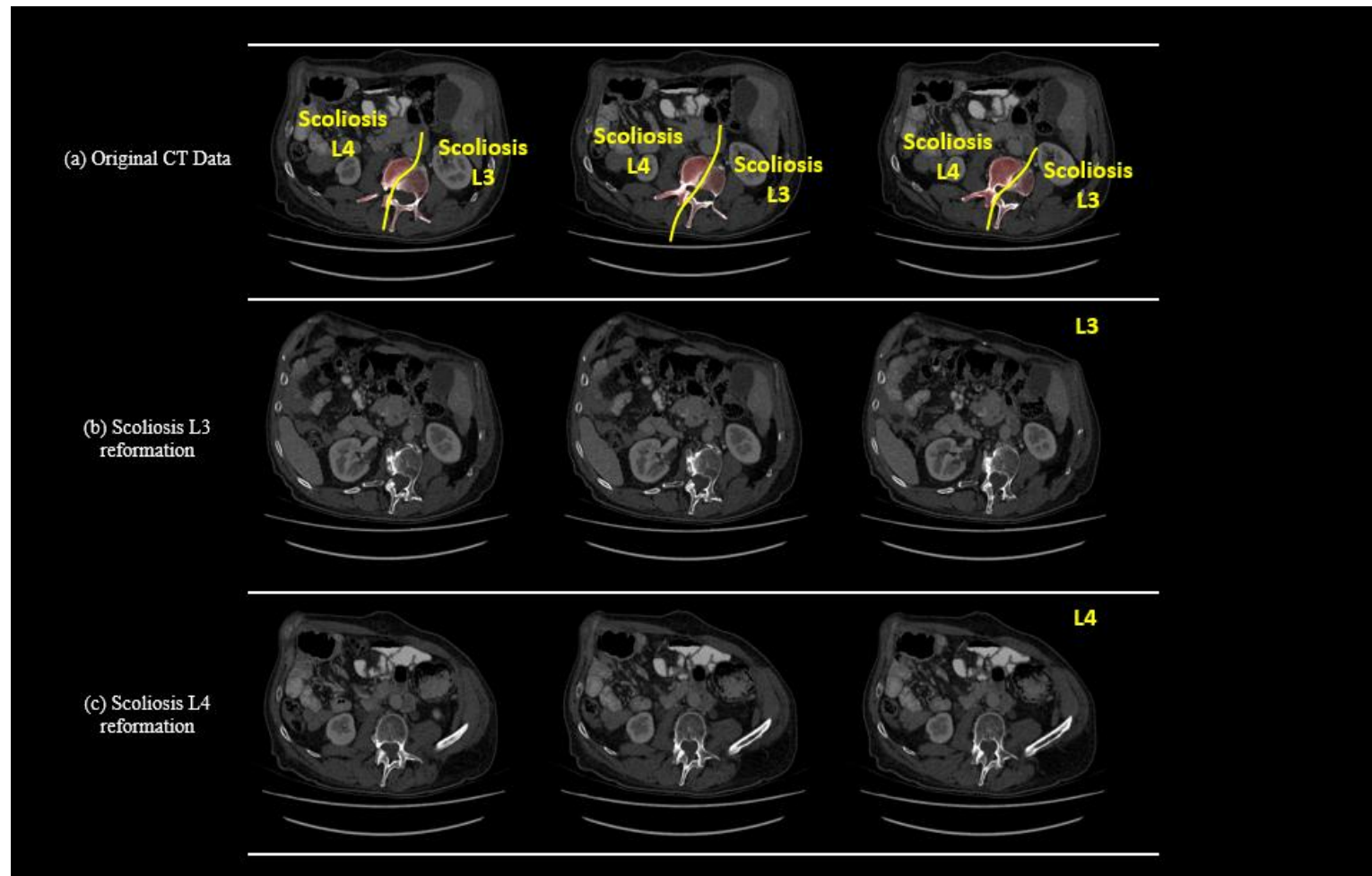


Fig. 2. This concept can also be applied to axial images. (a) shows the original CT data, illustrating the issues that arise when viewing a scoliosis patient in the axial plane. (b) and (c) present the results after reformation, where each slice clearly displays a single vertebral body.

Quantitative Evaluation

- Mild Scoliosis Group (Cobb Angle $\leq 25^\circ$): Comprised 24 patients with an average initial Cobb angle of 18.025° that was reduced to 3.049° after reformation, achieving a high correction rate of 83.085%.
- Moderate to Severe Scoliosis Group (Cobb Angle $> 25^\circ$): Consisted of 10 patients with an average initial Cobb angle of 28.676° that was reduced to 7.335° after reformation, demonstrating a correction rate of 74.421%.

Table 1. Average Cobb Angle Measurement According to Initial Cobb Angle Group

Group	Number of Cases	Avg Original Cobb Angle ($^\circ$)	Avg Reformatted Cobb Angle ($^\circ$)	Reduction (%)
Cobb Angle $\leq 25^\circ$	24 cases	18.025°	3.049°	83.085%
Cobb Angle $> 25^\circ$	10 cases	28.676°	7.335°	74.421%

- Instead of performing 3D reformation on the entire volume, it is necessary to choose a specific plane (e.g., sagittal or axial) and carry out the reformation process accordingly.
- Validation is required to assess how closely the reformed images resemble the original CT data.
- The performance of spine disease detection AI on the reformed images must be evaluated. If the AI recognizes the reformed images as unfamiliar or unseen data, additional training may be necessary.

- The proposed spinal reformation technique effectively corrects scoliosis-induced anatomical distortion in CT images.
- Anatomical normalization is expected to enhance the performance of AI-based diagnostic models.
- This method has the potential to improve clinical robustness in automated spine disease detection.

UNWAVERING ENDEAVORS

The 81st Korean Congress of Radiology and Annual
Delegate Meeting of the Korean Society of Radiology



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